AM2901 MANUAL TEST BOARD

DESIGN BASIS AND BILL OF MATERIALS

Single AM2901, manual controls, optional 4-bit SRAM data path

DESIGN INTENT

A slow, manually clocked learning and diagnostic board. DIP switches apply one AM2901 microinstruction at a time. A 62256 SRAM can supply D[3:0] during reads and can store Y[3:0] during writes. This design does not execute microcode automatically.

IMPORTANT LIMITS

This is a verified concept schematic, not a fabrication release. AM2901 variants and second-source parts may differ electrically. Confirm package orientation, pin names, supply current, and timing against the exact devices you own before ordering a PCB.

FUNCTIONAL PARTS

Qty	Reference	Part	Purpose
1	U1	AM2901B/C, DIP-40	4-bit ALU and 16 x 4 register file
1	U2	74HCT157, DIP-16	Select manual D[3:0] or SRAM DQ[3:0]
1	U3	74HCT125, DIP-14	Tri-state Y[3:0] write driver
1	U4	74HCT14, DIP-14	Debounce STEP and MEM WRITE buttons
1	U5	74HCT00, DIP-14	READ/WRITE decode and /WE gate
1	U6	74HCT244, DIP-20	Y and status LED buffer
1	U7	AS6C62256, DIP-28	32K x 8 SRAM; low nibble used
5	RN1-RN5	4.7 k pull-up networks	Defined switch levels; ON = logic 0
8	RLED1-RLED8	1.5 k	LED current limiting, about 2 mA
7	C1-C7	100 nF ceramic	One at every IC supply pair
1	C8	47 uF electrolytic	Board bulk decoupling

OPERATOR CONTROLS AND CONNECTORS

Qty	Reference	Item	Use
1	SWI	10-position DIP	I[8:0], one spare
1	SWA	4-position DIP	A[3:0] register address
1	SWB	4-position DIP	B[3:0] register address
1	SWD	4-position DIP	Manual D[3:0]
1	SWADDR	8-position DIP	SRAM A[7:0], 256 locations
1	SWMODE	DPDT ON-OFF-ON	MEM READ / OFF / MEM WRITE
2	SWSTEP/SWWR	Momentary NO	AM2901 clock and SRAM write
4	JSHIFT	3-pin headers/jumpers	RAM0/RAM3/Q0/Q3 shift endpoints
1	JPOWER	2-pin terminal	Regulated +5 V input
1	JEXP	2 x 10 header	Optional signal breakout/test

LOGIC CONVENTIONS

DIP switches connect signals to ground and use 4.7 k pull-ups; therefore switch ON means logic 0. Nets beginning with '/' are active-low. Use HCT, not HC, for signals driven by the bipolar AM2901. SRAM A[14:8] are fixed low in the base design, yielding 256 manually selectable locations while retaining the option to add address headers later.

RECOMMENDED FIRST BUILD

Populate the AM2901, switch banks, STEP circuit, and LED buffer first. Validate manual ALU/register operations before fitting the SRAM path. Sockets are strongly recommended for all DIP ICs, especially the AM2901.

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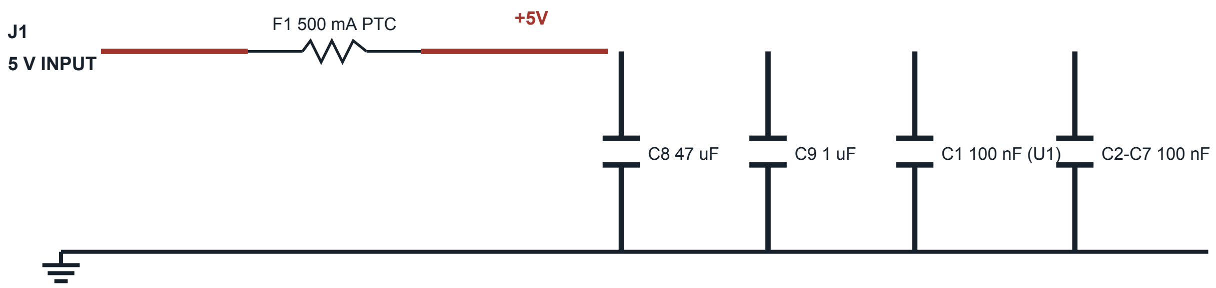
POWER, DECOUPLING, CLOCK AND MEMORY CONTROL

Exact values are suitable for slow manual operation

A. REGULATED 5 V POWER ENTRY



Place each 100 nF capacitor within a few millimetres of its IC.

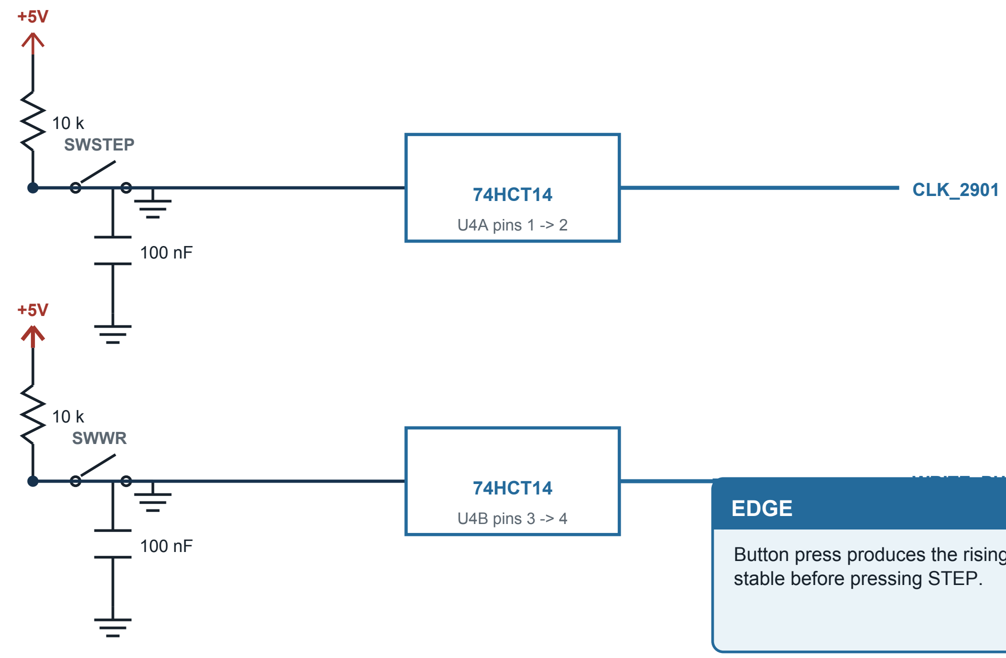
U1 AM2901: pin 10 = +5V; pins 11 and 30 = GND.

Use a regulated supply rated for at least 1 A during bring-up.

POWER CAUTION

Do not power the board from an unregulated wall adapter. Add reverse-polarity protection if the connector can be inserted incorrectly.

B. STEP AND WRITE BUTTON DEBOUNCE



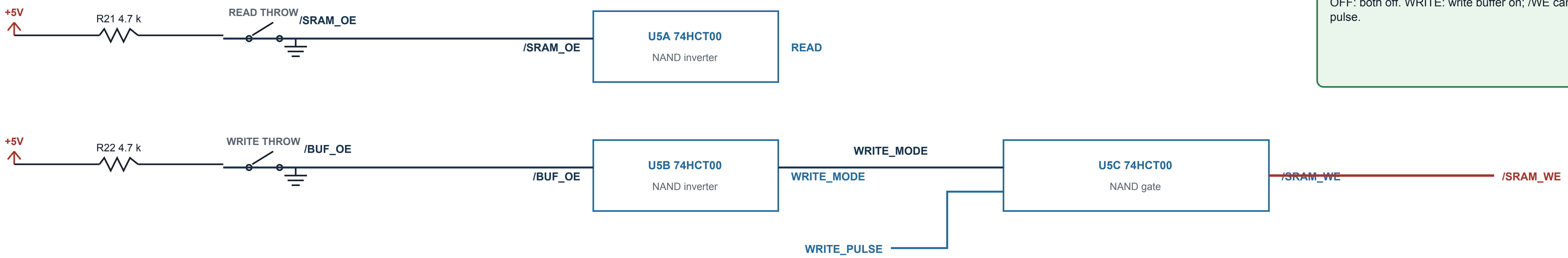
EDGE

Button press produces the rising edge. Register switches must be stable before pressing STEP.

C. MEMORY MODE SWITCH AND SAFE WRITE GATING

SWMODE: DPDT, center-off

READ position grounds /SRAM_OE. WRITE position grounds /BUF_OE.



TRUTH TABLE

READ: SRAM output enabled; write buffer off. OFF: both off. WRITE: write buffer on; /WE can pulse.

U5D unused NAND: tie both inputs to GND; leave output open.

Verify the exact parts in hand before fabrication.

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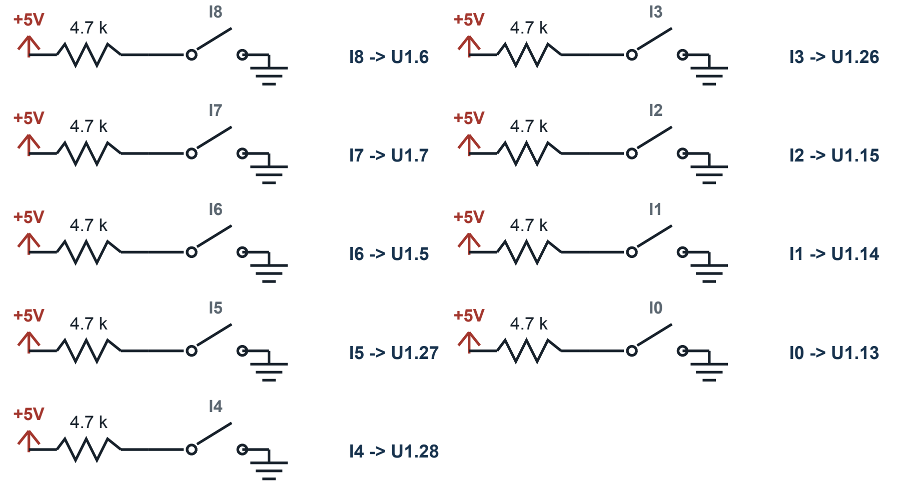
AM2901 CORE AND MANUAL CONTROL INPUTS

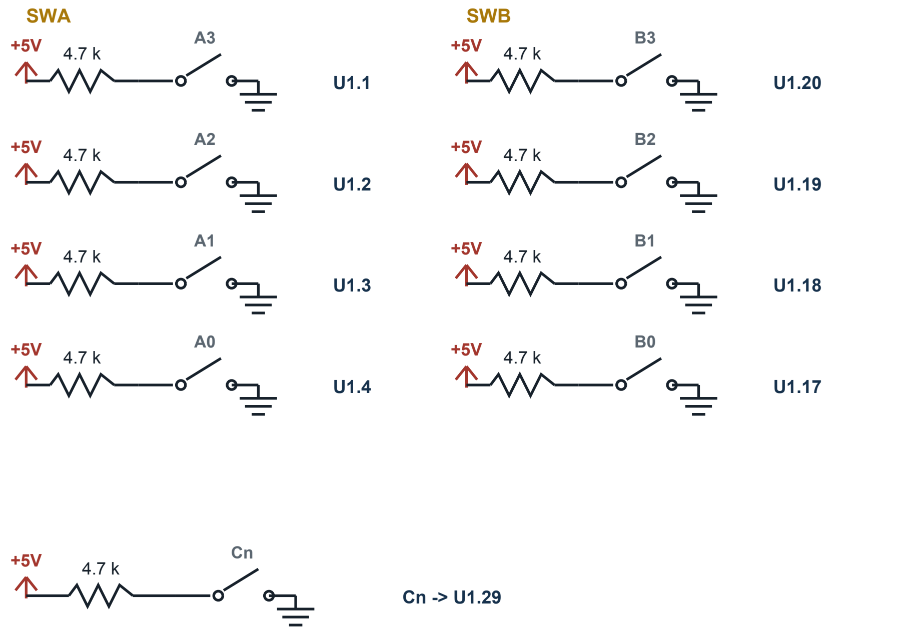
AM2901B/C P-40 or D-40 top-view pin assignment

INSTRUCTION SWITCH BANK SWI

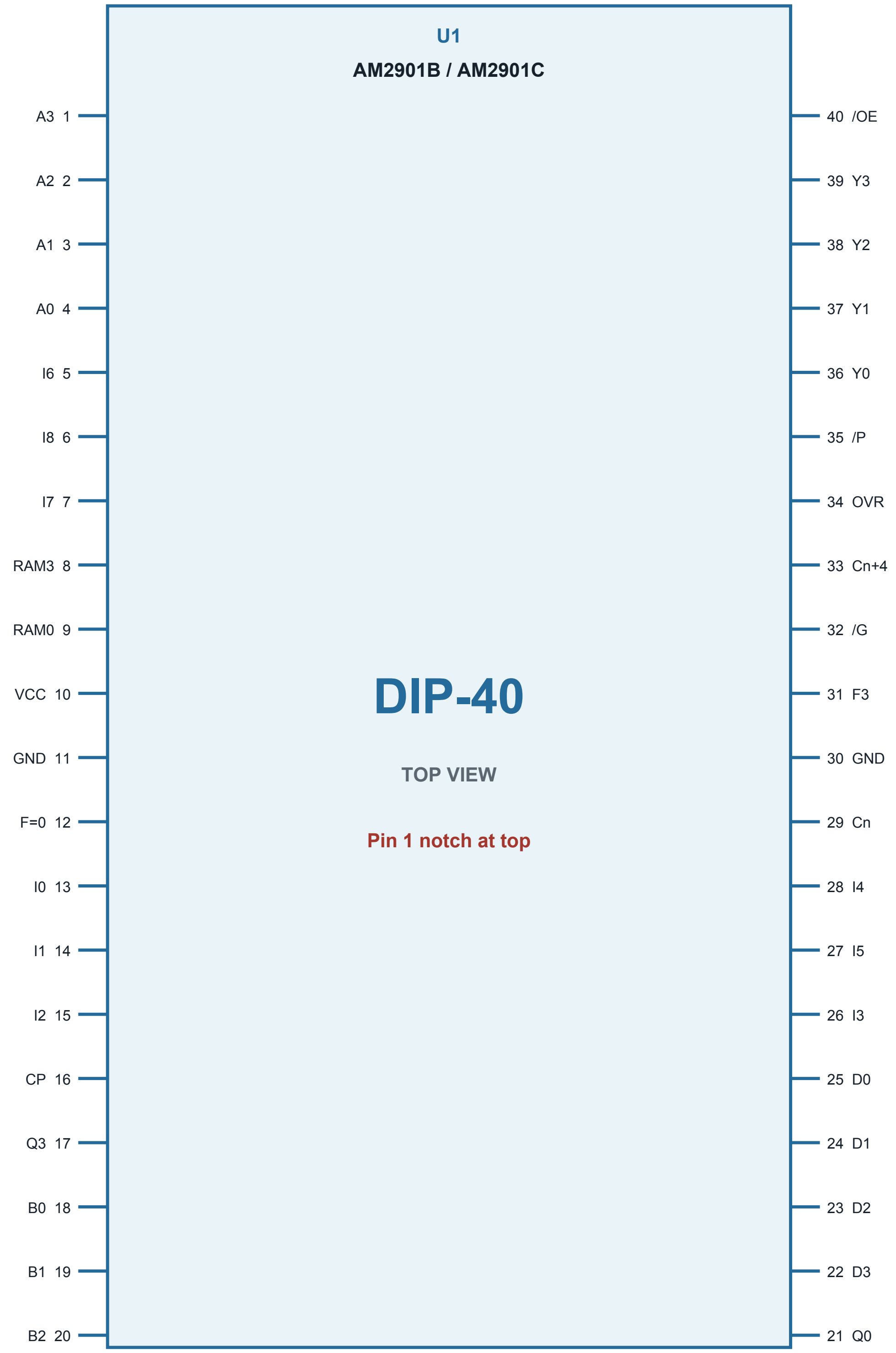


ON = 0. Label switch positions clearly on the silkscreen.

REGISTER ADDRESS, CARRY AND OUTPUT ENABLE



Tie U1.40 /OE LOW for continuous Y output.



CORE OUTPUT CONNECTIONS

- Y0 U1.36 -> U3.2 and U6 input
- Y1 U1.37 -> U3.5 and U6 input
- Y2 U1.38 -> U3.9 and U6 input
- Y3 U1.39 -> U3.12 and U6 input
- F=0 U1.12 -> U6 input
- F3 U1.31 -> U6 input
- OVR U1.34 -> U6 input
- Cn+4 U1.33 -> U6 input
- /P U1.35 and /G U1.32 -> test points

BIDIRECTIONAL SHIFT PINS

RAM0, RAM3, Q0 and Q3 can be inputs or outputs depending on I[8:6]. Do not hard-wire them directly to VCC or GND. Sheet 5 shows current-limited jumper circuits.

AM2901 POWER

U1 pin 10 to +5 V. U1 pins 11 and 30 both to the ground plane. Fit 100 nF from pin 10 to the nearest ground pin and an additional 1 uF nearby. The bipolar AM2901 may draw substantially more current than the CMOS support logic.

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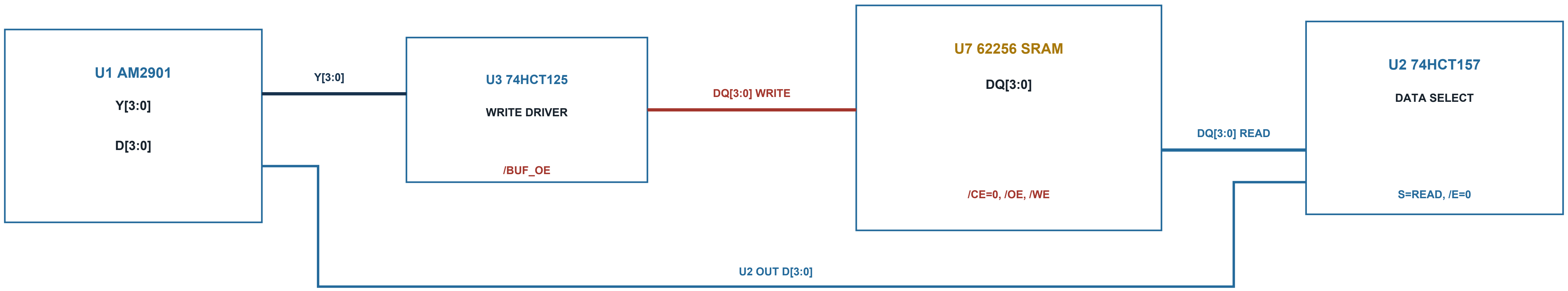
SRAM DATA PATH AND EXACT DIP CONNECTIONS

U2 selects read data; U3 isolates write data

A. FOUR-BIT MEMORY DATA PATH



B. U2 74HCT157 MAPPING

Bit	Manual I0	SRAM I1	Output to U1
0	U2.2 <- SWD0	U2.3 <- U7.11 DQ0	U2.4 -> U1.25 D0
1	U2.5 <- SWD1	U2.6 <- U7.12 DQ1	U2.7 -> U1.24 D1
2	U2.11 <- SWD2	U2.10 <- U7.13 DQ2	U2.9 -> U1.23 D2
3	U2.14 <- SWD3	U2.13 <- U7.15 DQ3	U2.12 -> U1.22 D3
Ctl	U2.15 /E = GND	U2.1 S = READ	U2.16 +5V; U2.8 GND

C. U3 74HCT125 AND U7 DATA MAPPING

Bit	AM2901 Y	74HCT125	SRAM DQ
0	U1.36 Y0	U3.2 A -> U3.3 Y	U7.11 DQ0
1	U1.37 Y1	U3.5 A -> U3.6 Y	U7.12 DQ1
2	U1.38 Y2	U3.9 A -> U3.8 Y	U7.13 DQ2
3	U1.39 Y3	U3.12 A -> U3.11 Y	U7.15 DQ3
Ctl		U3.1,4,10,13 = /BUF_OE	DQ4-DQ7 leave NC

WHY U3 IS USED

U1 /OE is tied low so Y remains continuously visible on the LEDs. U3 provides a separately controlled, tri-state copy of Y for SRAM writes. Only the WRITE position of SWMODE enables U3.

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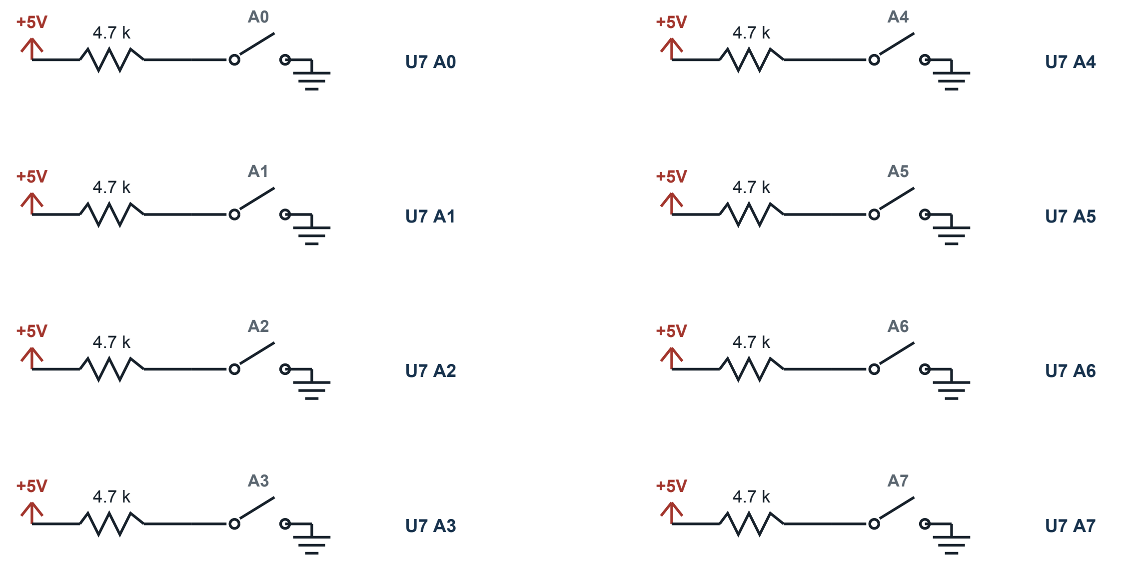
SRAM ADDRESSING, CONTROL PINS AND MEMORY OPERATION

Base configuration exposes 256 of 32768 locations

A. U7 AS6C62256 DIP-28 PIN CONNECTIONS

Pin	Signal	Connection	Pin	Signal	Connection
1	A14	10 k to GND / header	28	VCC	+5 V
2	A12	10 k to GND / header	27	/WE	U5C output
3	A7	SWADDR7	26	A13	10 k to GND / header
4	A6	SWADDR6	25	A8	10 k to GND / header
5	A5	SWADDR5	24	A9	10 k to GND / header
6	A4	SWADDR4	23	A11	10 k to GND / header
7	A3	SWADDR3	22	/OE	/SRAM_OE
8	A2	SWADDR2	21	A10	10 k to GND / header
9	A1	SWADDR1	20	/CE	GND
10	A0	SWADDR0	19-16	DQ7-4	No connect
11-13	DQ0-2	U2 + U3 data bus	15	DQ3	U2 + U3 data bus
14	GND	Ground plane			

B. ADDRESS DIP SWITCH CIRCUIT



Upper address pins A[14:8] are held low with individual 10 k resistors.

Optional 2 x 7 header pads allow later expansion without PCB changes.

C. MANUAL MEMORY OPERATING SEQUENCE

1. **READ**

   Set SWADDR. Set SWMODE to READ. U7 /OE goes low, U3 is disabled, and U2 selects SRAM DQ[3:0] onto AM2901 D[3:0]. Configure an AM2901 instruction that uses D as a source, then press STEP.

2. **WRITE**

   Set SWADDR. Configure the AM2901 so Y contains the desired value. Move SWMODE to WRITE; U3 now drives DQ[3:0]. Press MEM WRITE once to pulse U7 /WE low. Return SWMODE to OFF.

3. **VERIFY**

   Move SWMODE to READ at the same address and select an AM2901 operation that passes D to Y. The Y LEDs should reproduce the stored nibble.

WRITE DISCIPLINE

Do not change address or Y data while pressing MEM WRITE.

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LED DISPLAY, SHIFT ENDPOINTS AND TEST ACCESS

Buffers protect the AM2901 and improve observability

A. U6 74HCT244 LED BUFFER

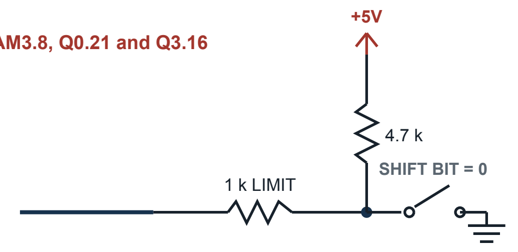
Displayed net	U6 input	U6 output	LED circuit
Y0	pin 2 (1A0)	pin 18 (1Y0)	1.5 k -> LED -> GND
Y1	pin 4 (1A1)	pin 16 (1Y1)	1.5 k -> LED -> GND
Y2	pin 6 (1A2)	pin 14 (1Y2)	1.5 k -> LED -> GND
Y3	pin 8 (1A3)	pin 12 (1Y3)	1.5 k -> LED -> GND
F=0	pin 11 (2A0)	pin 9 (2Y0)	1.5 k -> LED -> GND
F3	pin 13 (2A1)	pin 7 (2Y1)	1.5 k -> LED -> GND
OVR	pin 15 (2A2)	pin 5 (2Y2)	1.5 k -> LED -> GND
Cn+4	pin 17 (2A3)	pin 3 (2Y3)	1.5 k -> LED -> GND
Enable	pin 1 /1OE = GND	pin 19 /2OE = GND	Always enabled
Power	pin 20 = +5 V	pin 10 = GND	100 nF at pins 20/10

Each output: U6 output -> 1.5 k series resistor -> LED anode; LED cathode -> GND. Logical HIGH lights the LED.

B. BIDIRECTIONAL SHIFT ENDPOINT

Repeat for U1 RAM0.9, RAM3.8, Q0.21 and Q3.16

U1 SHIFT PIN



JSHIFT TEST NODE

USAGE

Open switch = weak HIGH for use as a serial input. Closed switch = LOW through 1 k. When the pin becomes an output, the 1 k resistor limits current if the switch state disagrees. Prefer leaving the switch open when observing an output.

Do not connect an LED directly to these pins.

C. TEST POINTS AND EXPANSION HEADER

Header pin	Net	Header pin	Net	Header pin	Net
1	+5V	8	D0	15	/SRAM_OE
2	GND	9	D1	16	/SRAM_WE
3	CLK_2901	10	D2	17	/BUF_OE
4	Y0	11	D3	18	F=0
5	Y1	12	Cn+4	19	F3
6	Y2	13	OVR	20	GND
7	Y3	14	READ		

PROBING

Provide individual loop test points for CLK_2901, /SRAM_WE, /SRAM_OE, /BUF_OE and all four DQ lines. These are the first signals to inspect if memory operation fails.

UNUSED INPUTS

Tie every unused HCT input to GND or +5 V. Never leave CMOS inputs floating. Unused outputs remain open.

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KICAD CAPTURE AND PCB LAYOUT CHECKLIST

Recommended net names, footprints and staged bring-up

A. KICAD SYMBOL / FOOTPRINT NOTES

- Create and verify a custom AM2901_DIP40 symbol from the pin table on Sheet 3.
- Use Package_DIP:DIP-40_W15.24mm for a standard 0.6-inch plastic DIP, but measure your part.
- Use DIP-28_W15.24mm for the 62256; HCT devices use 7.62 mm body-width DIP footprints.
- Assign pin 1 and notch markings prominently on silkscreen and assembly layers.
- Use resistor-network symbols whose common-pin orientation matches the purchased SIP networks.
- Add PWR_FLAG symbols only where KiCad ERC needs them; do not hide real power errors.

B. PCB PLACEMENT AND ROUTING

- Place U1 centrally with its 100 nF and 1 uF capacitors adjacent to pins 10, 11 and 30.
- Place U2 and U3 between U1 and U7 to keep D[3:0], Y[3:0] and DQ[3:0] short.
- Place switch banks along board edges in signal order; add unambiguous bit-number labels.
- Use a continuous ground plane. Avoid routing clock and /WE alongside long LED traces.
- Place SWMODE and SWWR apart from STEP to reduce operator mistakes.
- Keep all bus traces comfortably short; at manual speeds length matching is unnecessary.

C. RECOMMENDED KICAD NET NAMES

AM2901 CONTROL	I0 I1 I2 I3 I4 I5 I6 I7 I8 A0 A1 A2 A3 B0 B1 B2 B3 CIN CLK_2901
DATA PATH	D0 D1 D2 D3 Y0 Y1 Y2 Y3 DQ0 DQ1 DQ2 DQ3
STATUS	ZERO_N_OR_F0 NEG_F3 OVR COUT P_N G_N
MEMORY	MA0 MA1 MA2 MA3 MA4 MA5 MA6 MA7 SRAM_OE_N SRAM_WE_N BUF_OE_N READ WRITE_MODE WRITE_PULSE
SHIFT	RAM0_SHIFT RAM3_SHIFT Q0_SHIFT Q3_SHIFT

D. STAGED BRING-UP

1. Power only: check +5 V and current draw.
2. Fit HCT logic without U1/U7; verify STEP, mode and /WE waveforms.
3. Fit U1; test Y/status LEDs with manual operations.
4. Fit U7; write and read one address, then exercise all 256 locations.

PRIMARY REFERENCES

AMD Am2901B/Am2901C datasheet, connection diagram Figure 1. Alliance Memory AS6C62256 datasheet. Nexperia 74HC/HCT157, 74HC/HCT125, 74HC/HCT244 and 74HC/HCT14 datasheets.